

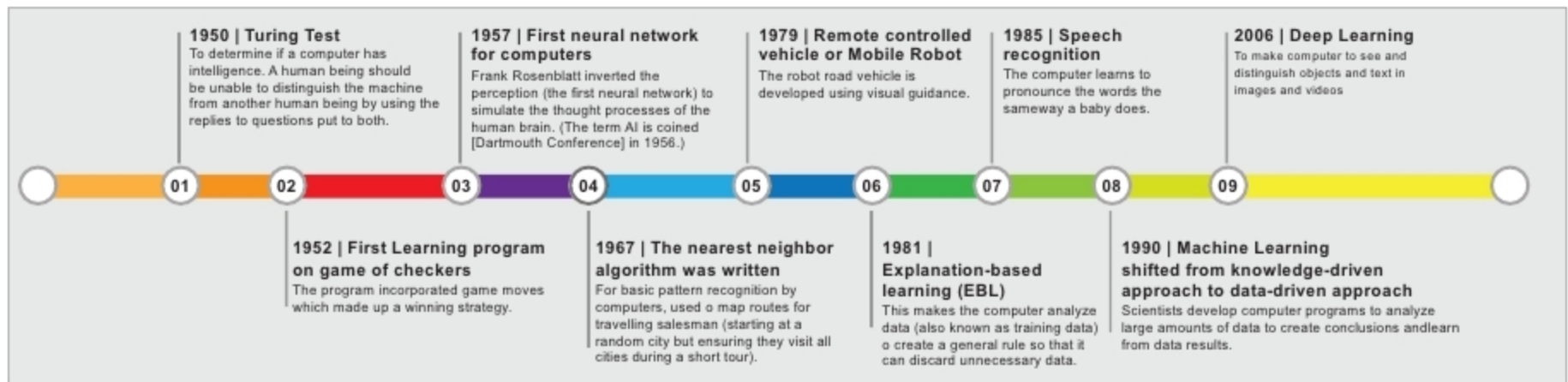


Figure 2: Evolution of machine learning

test samples are supplied to the model for evaluating its recognition capability. The following are some common examples where classification can be used.

1. A marketing manager analyses the data of customers to predict if they will buy a new cell-phone.
 2. A bank manager classifies the loan applicants as safe or risky.
 3. A space scientist classifies a spatial object as a planet, satellite or asteroid.
 4. A medical practitioner diagnoses a disease from past records.
- Typically, in classification, the categories (class label) are represented by discrete values, where ordering among values has no meaning.
 - Data classification is a two-step process. In the first step, a model is built by describing a predetermined set of data classes or concepts. In the second step, the model is used for classification and its accuracy is estimated. If the accuracy is acceptable, then the model is used to classify future data instances for which the classes are unknown.
 - Some commonly used classifiers include the decision tree, artificial neural networks and support vector machines.

Unsupervised learning: In this case, the machine is equipped with examples but not classes; it is somewhat an exploratory analysis. This is learning from observation.

- **Clustering** is an unsupervised learning approach for organising objects into groups whose members are similar in some way.
- Objects in a cluster possess the principle of maximum intra-cluster similarity and minimum inter-cluster similarity.
- Different algorithms use different measures for similarity between objects to group them together. In clustering, learning takes place through observation rather than training.
- Some applications of clustering include anomaly or fraud detection and speaker diarisation.

Reinforcement learning: An algorithm is given a penalty or a reward for successfully completing a task or failing to do it, respectively. This can be a small task of a robot identifying a hurdle or an obstacle in front of it. This is incentive based learning.

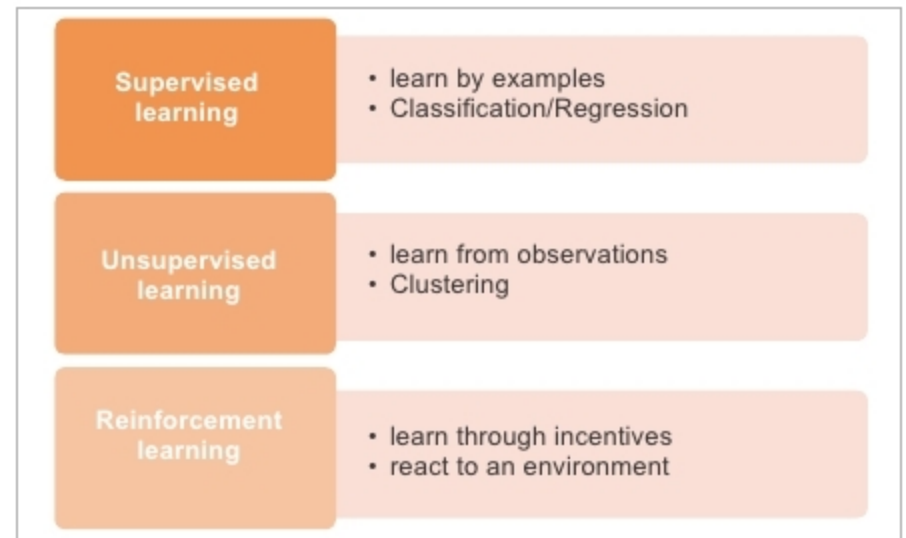


Figure 3: Machine learning approaches

ML based problem-solving approaches

A conventional pipeline for an ML application is depicted in Figure 4. It consists of the following four stages.

- **Data preparation:** Input data is usually pre-processed, and then fed into the next module for further processing.
- **Feature extraction:** Features are quantitative variables which best represent the input data. Several algorithms for feature extraction are available in research literature, specific to different domains.
- **Classifier:** A typical task of ML is classification. It goes through the data instances and classifies them into one or another class after building the classifier model using training data.
- **Prediction and evaluation:** A classifier predicts the class label of a given instance on an unknown sample. The prediction performance is measured for correctness, using performance metrics such as accuracy.

An illustration

Let's illustrate this pipeline using the specific application of speaker verification. In this system, a user wants to access an online banking system remotely using his voice as the password. During the enrolment stage, all authentic users are registered with the systems by providing their voice samples. These samples are used to build models for each user. During the authentication stage, a registered user provides his voice sample to the system, which is matched with the stored voice templates of that user. If a match is found successful, the user is accepted as genuine; else his access is rejected.